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Effect of Airfoil-Preserved Undulation Placement on Free Shear Layer

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Abstract

The distribution of mean and fluctuating quantities in the free shear layer wake of airfoil-preserved undulated wings are investigated using Particle Image Velocimetry (PIV) at the University of Dayton Low Speed Wind Tunnel (UD-LSWT). Prior work on wings with airfoil-preserved undulations revealed that wings with trailing edge undulations and wings with leading and trailing edge undulations had higher aerodynamic efficiency than the wings with leading edge undulations. The flow conditions that exist around the undulated wings, which brings about differences in the aerodynamic efficiency, are investigated by quantifying the near wake mean velocity and vorticity along with turbulent properties such as Reynolds stress and turbulence intensity at different angles of attack. Across all the wake properties, it was observed that the trailing edge undulated wings had a lower Reynolds stress and turbulence intensity magnitude before 12° when compared to any other cases including the baseline. The wings with both leading and trailing edge undulations showed characteristics of the leading edge-only and trailing edge-only undulated wings.

Nomenclature

α	=	angle of attack (degrees)
λ	=	wavelength (cm)
ω_z	=	z-vorticity (1/s)
A	=	amplitude (cm)
c	=	chord (cm)
LE	=	leading edge
TE	=	trailing edge
LETE	=	leading and trailing edge
TI	=	turbulence intensity (m/s)
u	=	mean velocity in x-direction
u'	=	fluctuating velocity component in x-direction
v	=	mean velocity in y-direction
v'	=	fluctuating velocity component in y-direction

I. Introduction

Wings with undulations have been investigated for use in multiple applications such as aircraft wings [1], [2], [3], [4], wind turbine blades [5], aircraft propeller blades [6], marine propeller blades [7], electronic heat sink fans, etc. to improve aerodynamic efficiency [2], delay stall [1], [3], [4] and reduce noise [8]. Most of these investigations include leading edge undulations while very few incorporate trailing edge undulations and undulations on both leading and trailing edges [2]. Numerous studies have found that an increase in undulation amplitude and decrease in undulation wavelength was found to improve the aerodynamic efficiency of wings with leading edge undulations [4], [9], [10], [11], [12]. One such airfoil-preserved undulations study was done by Loughnane et al. [13], where the aerodynamic performance of wings with airfoil-preserved undulations with NACA 0012 cross-sections along the leading edge, trailing edge, and both the leading and trailing edge were investigated. The wavelength of the undulations was varied to create 6, 9, and 12 undulations along the semi-span model of the wing with the wavelength-to-chord ratio (λ/c) of 0.15, 0.21 and 0.31 respectively. Fig. 1 below from [13] shows the variation of aerodynamic efficiency (C_L/C_D) for all cases tested with respect to angle of attack (Fig. 1a) and the percent difference in the

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aerodynamic efficiency when compared to the baseline with angle of attack (Fig. 1b). Data points above the horizontal axis in Fig. 1b would indicate an increment in aerodynamic efficiency when compared to the baseline. The wings with leading edge undulations indicate the same trends as seen in the literature where higher efficiency is seen with a decrease in undulation wavelength (number of undulations). However, the trailing edge undulations show a different trend than that of leading edge and both leading and trailing edge undulations. The wings with trailing edge undulations outperform the other undulated wing cases, especially at lower angles of attack. Interestingly, a genetic optimization study conducted by Lohry et al. [14] on undulated wings found that the optimized geometry in terms of maximum aerodynamic efficiency (L/D) is a wing with undulations along the trailing edge as shown in Fig. 2. This observation in both the experimental study [13] and numerical simulation [14] conclusively show that even though the wings with leading edge undulations help delay stall (just like vortex generators), the aerodynamic efficiency cannot be maximized.

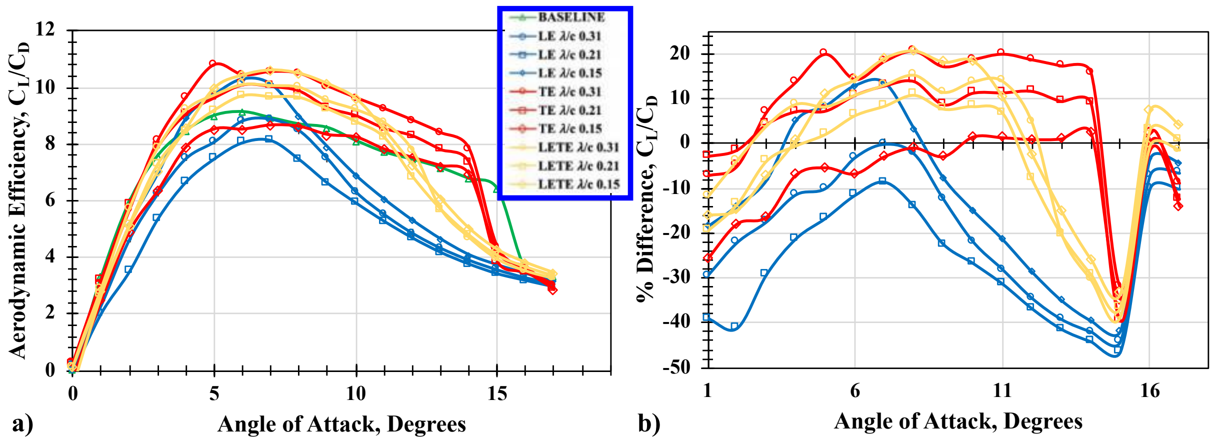

Fig. 1 Variation of C_L/ C_D a) comparison of all cases and b) percent difference from baseline NACA 0012 wing [13]

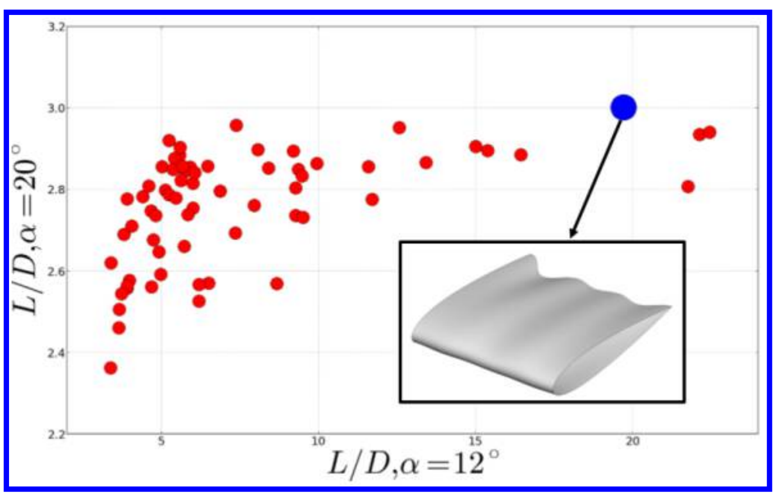

Fig. 2 Two-objective genetic optimization based on L/D and potential optimal solution, adapted from [14]

Free shear layer investigations conducted by Gunasekaran et al. [15] and Stevens et al. [16] showed evidence that the near and far wake properties are directly related to the aerodynamic efficiency of the wing. Investigations by Yarusevych [17] and Guan [18] also found that the near wake properties are influenced by the Reynolds number and trailing edge geometry. Therefore, by investigating the free shear layer in the near wake of the undulated wings, strong correlations can be drawn between the force and the flow field data. The flow around the wings with undulations is quite complex due to the formation of laminar separation bubble and streamwise counter-rotating vortices. New et al. [19] conducted a PIV study to investigate the flow separation behavior associated with a hydrofoil featuring leading edge tubercles. The study found that while the baseline hydrofoil and the trough plane of the tubercled hydrofoil exhibited relatively consistent flow separation, the flow separation observed along the peak plane of a tubercled hydrofoil shows fluctuations at 15° angle of attack. Figure 3 below shows the observed results for the baseline hydrofoil along with the tubercled hydrofoil along both the peak (crest) plane and trough plane. This shows that while the baseline wing exhibits a distinct flow separation bubble along the upper surface, the peak plane of the tubercled airfoil exhibits flow that does not detach from the leading edge while the trough plane exhibits a leading edge separation bubble followed by the flow reattaching to the surface. It is not clear whether same physics will be observed on wings with trailing edge undulations or wings with both trailing and leading edge undulations at lower angles attack.

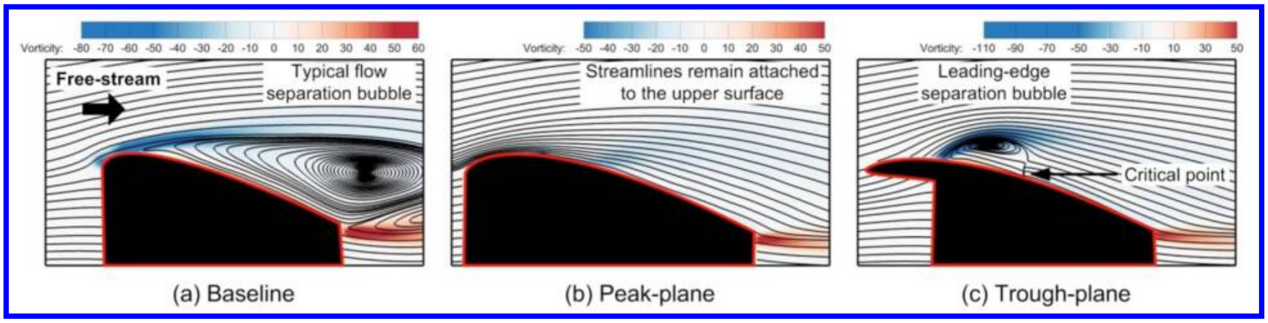

Fig. 3 Mean velocity and vorticity distributions for baseline and tubercled hydrofoils, adapted from [19]

Therefore, the motivation to investigate the free shear layer (FSL) wake of airfoil preserved undulated wings is two-fold: 1) validate the differences in the aerodynamic coefficients of the leading edge, trailing edge and both leading and trailing edge undulations by investigating the near wake mean and turbulent flow quantities. 2) identify the self-similarity (if any) in the mean and fluctuating properties of the FSL across the undulations for wings with different undulation locations (leading edge, trailing edge and leading and trailing edges). The wings with the highest aerodynamic efficiency was chosen from the force-based testing data for each undulation placement cases (TE λ/c 0.31, LE λ/c 0.15, LETE λ/c 0.15) for this study.

II. Experimental Setup

A. Wind Tunnel

All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) in the open jet configuration (Fig. 4). The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens and 4 interchangeable 76.2cm x 76.2cm x 243.8cm (30” x 30” x 96”) test sections. A photo of the UD-LSWT open jet configuration is shown in Fig. 4. The test section is convertible from a closed jet configuration to an open jet configuration with the freestream range of 6.7m/s (20 ft/s) to 40m/s (140 ft/s) at a freestream turbulence intensity below 0.1% measured by hot-wire anemometer. The contraction feeds into a pressure/air sealed room where the test section is located. The effective length of the test section in the open jet configuration is 182cm (72”). A 137cm x 137cm (44” x 44”) collector collects the expanded air on its return to the diffuser.

B. Test Model Design and Fabrication

Three undulated wing cases (one with trailing edge undulations, one with leading edge undulations and one with both leading and trailing edge undulations) were considered for the current study. The undulated wing cases along with the baseline NACA 0012 wing are depicted in Fig. 5 for clarity. All undulated wings have a maximum chord length of 12.70 cm (5 in), a minimum chord length of 10.16 cm (4 in) and a wingspan of 25.40 cm (10 in) resulting in an effective full span AR of 4.5. In comparison, the baseline NACA 0012 wing has a consistent chord length of 12.70 cm (5 in) and a wingspan of 25.40 cm (10 in) resulting in a full span AR of 4. Table 1 lists the wavelength, amplitude, surface area, and planform area measurements for each undulated wing under consideration. The wavelength is defined by the width of each undulation and determines the number of undulations along the span. The amplitude is defined by the chordwise depth of the undulations and is provided where applicable (at the leading edge, trailing edge, or both leading and trailing edges.) The naming convention for the undulated wing conventions begins with the placements of the undulations followed by the wavelength of the undulations normalized by the chord length. A NACA 0012 airfoil was maintained throughout the span for all wing cases.

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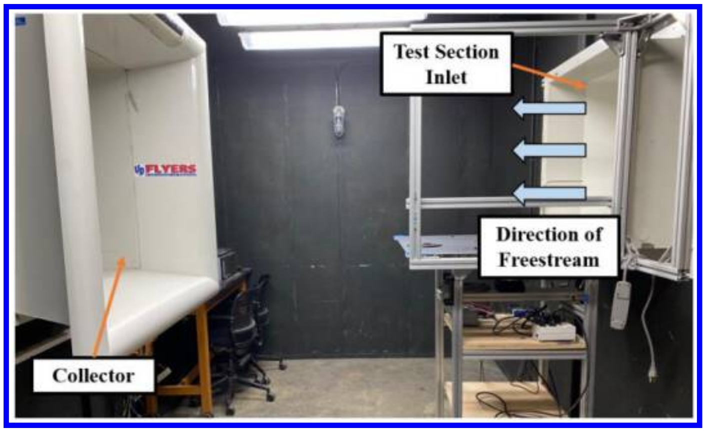


Fig. 4 UD-LSWT in open-jet configuration

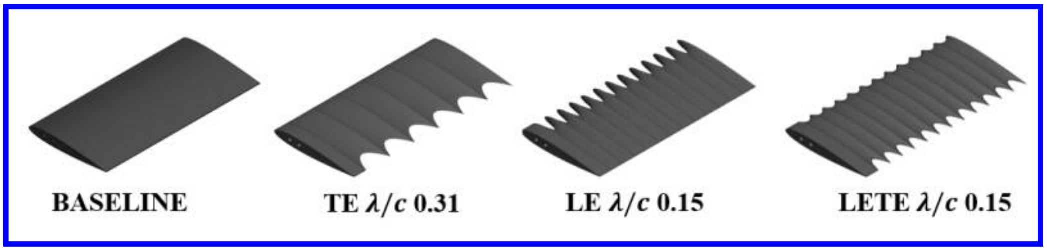
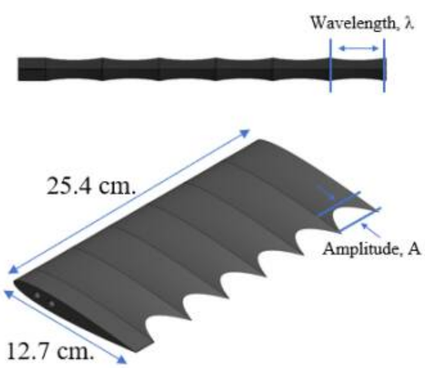


Fig. 5 Schematic of wing models with varying undulation numbers and locations

Table 1 Dimensions for undulated wings



Case	A, cm	λ, cm	Surface Area, cm²	Planform Area, cm²
TE λ/c 0.31	2.54	3.91	308.23	285.35
LE λ/c 0.15	2.54	1.96	328.00	
LETE λ/c 0.15	0.635 (LE) 1.905 (TE)	1.96	314.32	

All the wing models shown in Fig. 5 were generated in SolidWorks. To generate the TE undulations, the leading-edge of the airfoil profile was kept at a constant location and the trailing edge of the wing was reduced by 80% at the center of each undulation. A similar procedure was undertaken to create the LE undulations. The wing with both LE and TE undulations was modeled by placing the aerodynamic center of the two airfoil profiles at the same location. The loft command in SolidWorks was used to preserve the airfoil in all the wing cases. The reduction in chord length in center of each undulation leads to a reduction in airfoil thickness along the span of the wing when the two profiles are lofted. The loft was then mirrored to produce the desired number of undulations. All wings were 3D printed using the University of Dayton Gorilla Maker 3D printer.

C. Streamwise PIV Test Setup

Streamwise Particle Image Velocimetry (PIV) was conducted to analyze how the undulations affected the free shear layer compared to the straight NACA 0012 wing. A schematic for the streamwise PIV experimental setup can be seen in Fig. 6 below. The setup in the UD-LSWT for the baseline wing can subsequently be seen in Fig. 7. The

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PIV experiment was conducted using a Vicount smoke seeder with glycerin oil, a 200 mJ/pulse Nd:YAG frequency doubled laser (Quantel Twins CFR 300) and an Imperx B2021 camera with 200 mm lens**.** The time delay between the laser pulses was set to obtain 8 to 10-pixel particle displacement in the free shear layer. A plano-convex and plano-concave lens were used in series to open the laser beam into a sheet. The laser and camera were triggered simultaneously using a Quantum composer pulse generator. In each test case, 1,500 image pairs were obtained and processed using ISSI Digital Particle Image Velocimetry (DPIV) software. Two iterations were performed in DPIV processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration.

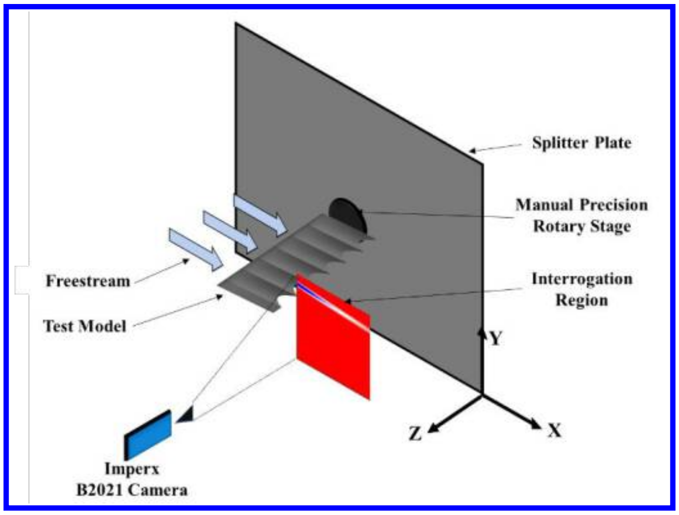

Fig. 6 Schematic of PIV experiments and equipment used in UD-LSWT

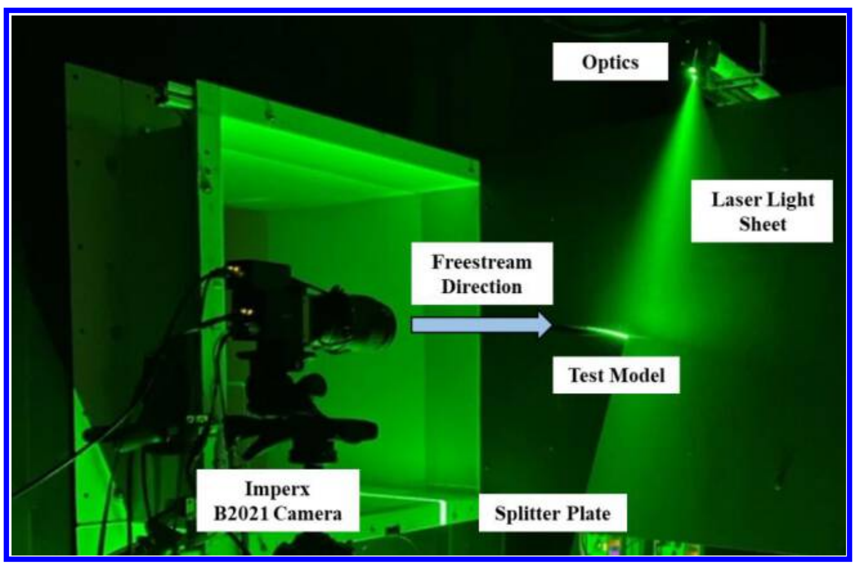

Fig. 7 Experimental setup in UD-LSWT, baseline case shown

For the undulated wings, data was collected for two different laser sheet placements, one plane aligning with a cross-section of the maximum chord of the undulated wing (peak plane) and the other plane aligning with a cross-section of the minimum chord of the undulated wing (trough plane). A schematic illustrating the two laser sheet placements for the LE λ/c 0.15 case is shown in Fig. 8 below.

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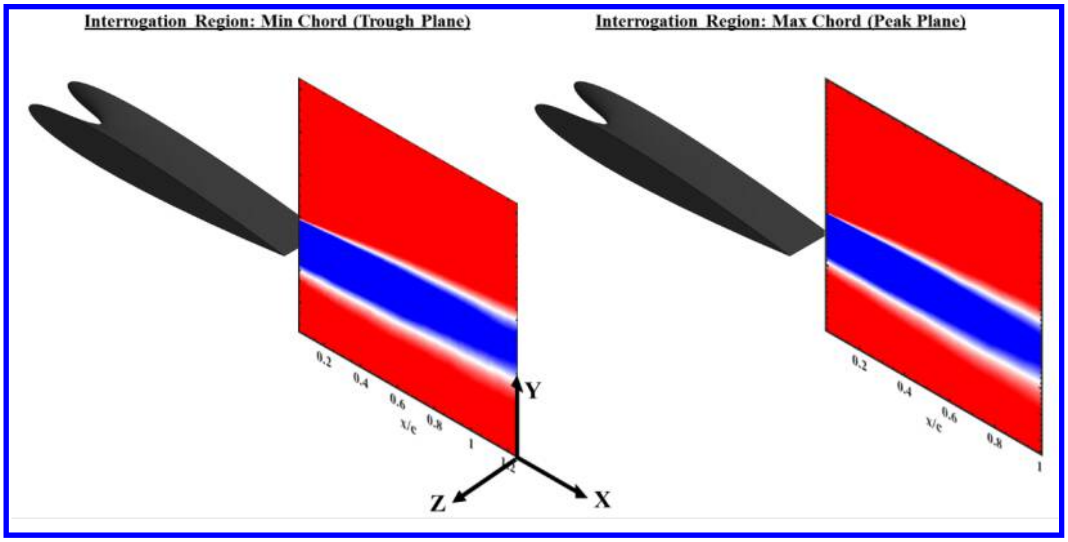

Fig. 8 Schematic of interrogation regions at trough plane and peak plane

Table 2 shows the test matrix for the streamwise PIV experiment. The streamwise PIV experiment was conducted at a Reynolds number of 223,481 for the baseline case and peak plane of the undulated wings. The Reynolds number for the trough plane of the undulated wings was is inherently lower (178,785) as the chord length at the trough plane is 80% of the chord length at the peak plane.

Table 2. Test Matrix for Streamwise PIV Experiment

Case	Field of View Size (m)	Spatial Resolution (pix/m)	Local Reynolds Number	Laser Time Delay (μs)	Size, m x m	Freestream Speed, m/s	AOA Range, °	AOA Increment, °
TE λ/c 0.31 TROUGH	0.127	16,315	178,785	29	0.127 x 0.254	25	2 to 12	2
TE λ/c 0.31 PEAK	0.123	16,846	223,481					
LE λ/c 0.15 TROUGH	0.133	15,579	178,785					
LE λ/c 0.15 PEAK	0.134	15,463	223,481					
LETE λ/c 0.15 TROUGH	0.133	15,579	178,785					
LETE λ/c 0.15 PEAK	0.129	16,062	223,481					
BASELINE	0.124	16,710	223,481					

III.Results

A. Mean Flow Properties

1. Mean Velocity

The near wake mean velocity contours are shown in Fig. 9 below for all the cases tested. Two sets of contours are presented for each undulated wing case, one corresponding to the trough plane (minimum chord) and the other corresponding to the peak plane (maximum chord) as labeled. As expected, all the cases show an increment in wake thickness with increase in angle of attack. It is also seen for the TE and LETE undulated wings that the wake thickness in the trough plane is thinner when compared to the wake thickness in the peak plane. This could be due to the variation in the local Reynolds number, but the different wake thickness for the same trough Reynolds number for different undulated cases shown in Fig. 9 indicates that the wake thickness is also a function of undulation placement.

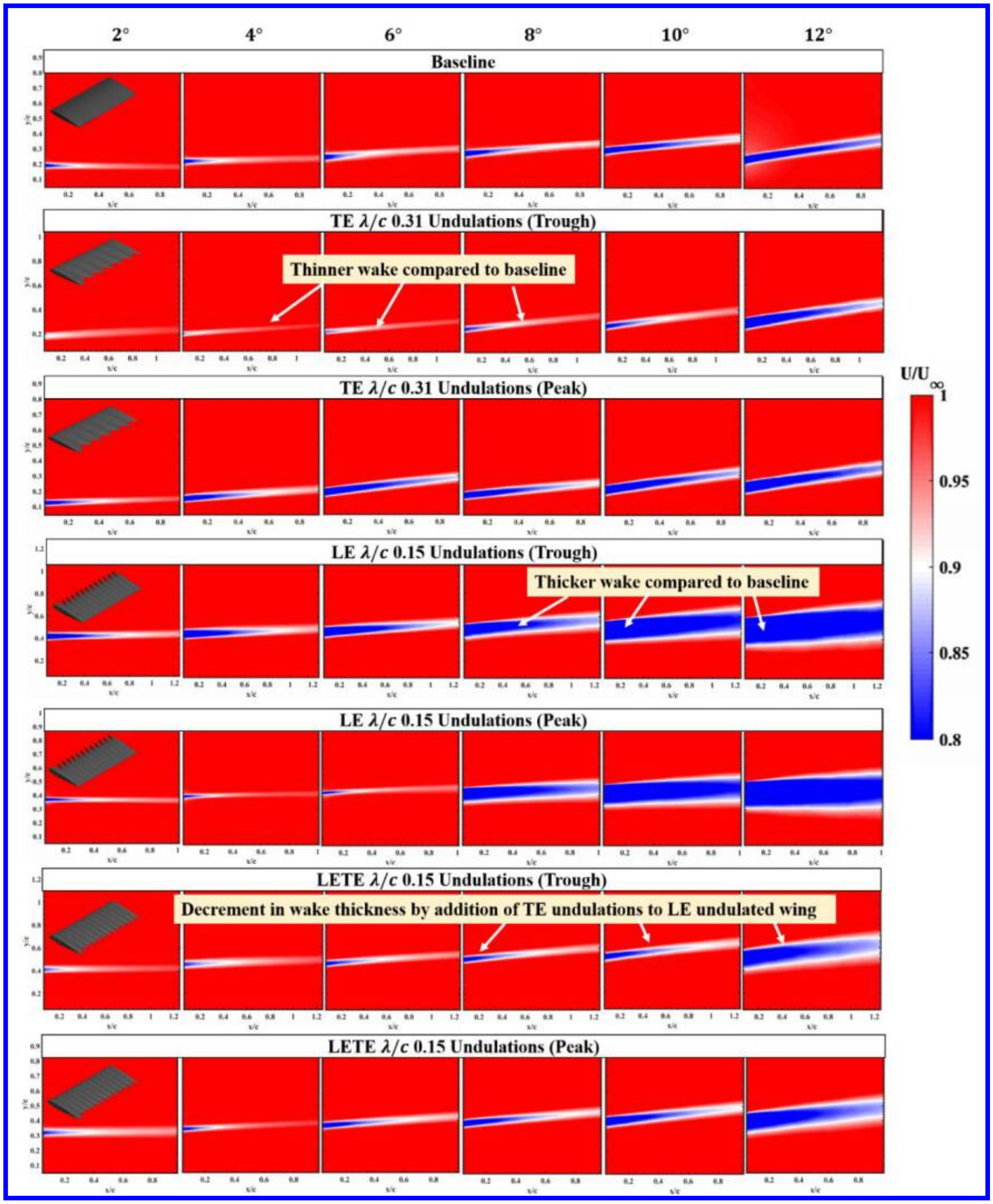

Fig. 9 Mean normalized u-velocity contours with increasing angle of attack for baseline, TE undulated wing, LE undulated wing, and LETE undulated wing

The trough plane of the TE undulated wing consistently produces a thinner wake and momentum deficit of less magnitude when compared to the baseline as well as any other undulated wing cases. Since the wake thickness, combined with the momentum deficit magnitude can be directly related to the drag coefficient of the wing, the trend seen in Fig. 9 for the TE undulated wing matches well with the coefficient of drag variation shown in Loughnane et al. [13] where the wing showed consistently lower drag coefficient than other cases, including the baseline at angles

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of attack greater than 8°. It is also observed that the peak plane of the TE undulated wing case has a higher wake thickness and the momentum deficit magnitude when compared to the baseline. This could be attributed to the three-dimensional effect created by the undulated wings where a horse-shoe type vortex might form as observed in wings with TE serrations [20].

The wing with LE undulations shows a slight increment in wake thickness and momentum deficit magnitude as the TE undulation until 8° angle of attack. Beyond 8° however, a significant increase in wake thickness was observed, indicating the onset of flow separation. This has a detrimental performance in drag coefficient where the magnitude increases significantly with increase in angle of attack when compared to the baseline. There are many reported cases in the literature where the leading edge undulations help delay stall. While this is true, at lower angles of attack, the leading edge undulations also significantly increase the drag coefficient thereby decreasing the aerodynamic efficiency.

The variation of aerodynamic coefficients for different undulated wings documented in [13] (also in Fig. 1) indicate that the wings with leading and trailing edge undulations show performance characteristics similar to both the trailing edge-only and leading edge-only undulated wings. This trend can be seen once again in the near wake thickness and the momentum deficit magnitude where the wake thickness and the momentum deficit magnitude are higher than the TE case but lower than the LE case. The changes in the near wake properties upon addition of the trailing edge undulations to the leading edge undulations are apparent. It should be noted that the local Reynolds number for all three cases in the trough plane are the same. The wavelength to chord ratio in both the LE and the LETE case is also the same. Therefore, the differences in the near wake properties indicates the benefits of having trailing edge undulations along with the leading edge undulations rather than having leading edge undulations alone. It was shown in [13] that the LETE case would offer both the benefits of TE and LE case where the drag coefficient is reduced at lower angles of attack and help delay stall at higher angles of attack. The near wake contours shown in Fig. 9 help make that case once again.

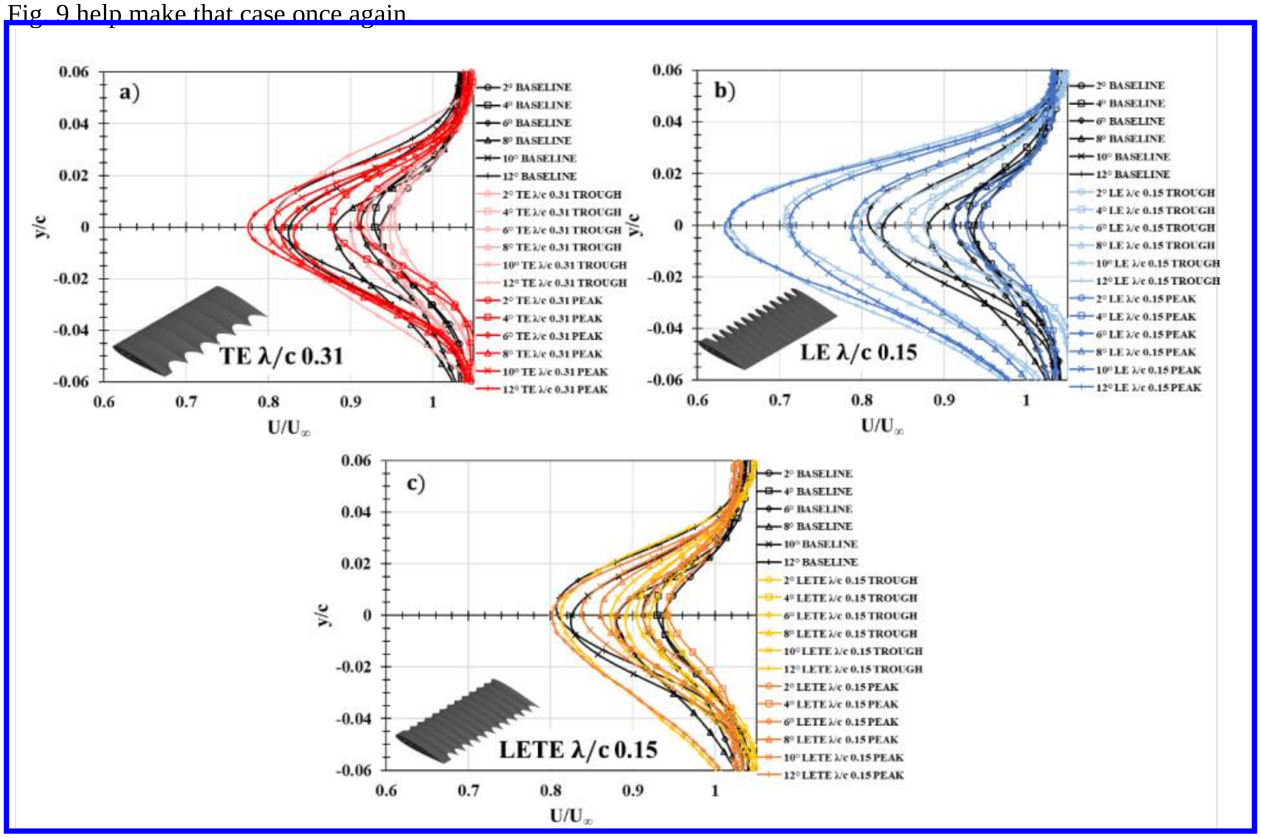

Fig. 10 Mean normalized u-velocity profiles with increasing angle of attack for a) TE λ/c 0.31 case compared to baseline, b) LE λ/c 0.15 case compared to baseline and c) LETE λ/c 0.15 case compared to baseline

The momentum deficit profiles obtained from the contours in Fig. 9 are shown in Fig. 10. These profiles were obtained by averaging 5-10 columns of data in the middle of the field of view. Figure 10a shows the momentum deficit profiles for both the trough and peak planes of the TE wing compared to the NACA 0012 baseline case. The momentum deficit profiles for both peak and trough planes is comparable to the baseline with the trough plane showing a lower momentum deficit magnitude when compared to the baseline. Figure 10b shows the momentum deficit profiles for both the trough and peak planes of the LE wing compared to the NACA 0012 baseline case. It is evident from Fig. 10b that the momentum deficit behind the trough plane of the LE wing is higher than that of the baseline at all angles of attack, especially after 8° angle of attack. Figure 10c shows the momentum deficit profiles for both the trough and peak planes of the LETE wing compared to the baseline. Out of all three cases, it can be seen that the momentum deficit profiles of the LETE case matches well with that of the baseline.

2. Vorticity

The Z-vorticity contours and profiles behind the wake of all cases can be seen in Figs. 11 and 12 below. The Z-vorticity in the wake was determined using Eqn. 1 below.

$$\omega_z = \left(\frac{\partial V}{\partial x} - \frac{\partial U}{\partial y}\right) \tag{1}$$

Figure 11 shows increasing vorticity magnitude with increasing angle of attack for both the baseline case and the trough and peak planes of the TE case. For the LE and LETE cases, however, the peak vorticity magnitude generally decreases with increasing angle of attack, and the vorticity is distributed over a larger area of the wake, especially behind the top surface of the wing. This larger spread in vorticity is a sign of flow separation where the vorticity is distributed across a wider area of the wake instead of being concentrated in the shear layer. The magnitude of near wake vorticity is decreased when compared to the baseline in the trough plane of the TE undulated wing when compared to the baseline. However, in the peak plane, the vorticity magnitude is seen to increase when compared to the baseline. Once again, this could be due to the three-dimensional horse-shoe vortex development at the trailing edge at the peak plane. The opposite behavior is seen in the LE case. The vorticity magnitude is higher in the trough plane when compared to the peak plane. This could be due to the formation of secondary flow in the trough of the undulation where a pair of streamwise counter rotating vortices is known to occur on wings with leading edge undulations [21].

Similar to the trend observed in the momentum deficit, the vorticity is distributed over a large area of the wake after 8° angle of attack in both upper and lower surfaces. It is suspected that the near wake of the LE undulated wing is highly three dimensional. The vorticity magnitude in the near wake of LETE undulated wing shows peculiar behavior where vorticity magnitude remains almost consistent as a function of angle of attack, except for at higher angles of attack. The distribution of vorticity to a larger area in the wake is pushed to 12° angle of attack when compared to the wings with LE undulations alone where the larger spread in vorticity distribution starts at 8° angle of attack. Therefore, the addition of the trailing edge undulation to the leading edge undulated wing brings about drastic changes in both the mean velocity and vorticity where the abrupt flow separation is reduced a more gradual increment in wake thickness is observed. The LETE undulated wings once again shows benefits at lower and higher angles of attack when compared their counterparts where the advantages of the undulations are only seen in either lower or higher angles of attack.

The vorticity profiles shown in Fig. 12 help show the trends seen in contours. It can be seen that the undulated wing cases reduce the vorticity magnitude in the wake of the wing when compared to the baseline. The TE undulated wing follows closely to the trends seen in the baseline but never exceed the peak vorticity magnitude in any of the angle of attack cases. The distribution of the vorticity magnitude from the upper and lower surface is also similar to that of the baseline. Figure 12 also helps the spread of vorticity out over a wider area in LE and LETE cases at high angle of attack.

B. Fluctuating Flow Properties

Apart from quantifying the differences in the mean wake properties among the different undulated cases, it is also important to determine the variations in the fluctuating velocity components and properties to understand the effect of undulations on the wake turbulence. The discussion on the variation in wake turbulent intensity and Reynolds stress is included in this section, along with the discussions on two-point correlations.

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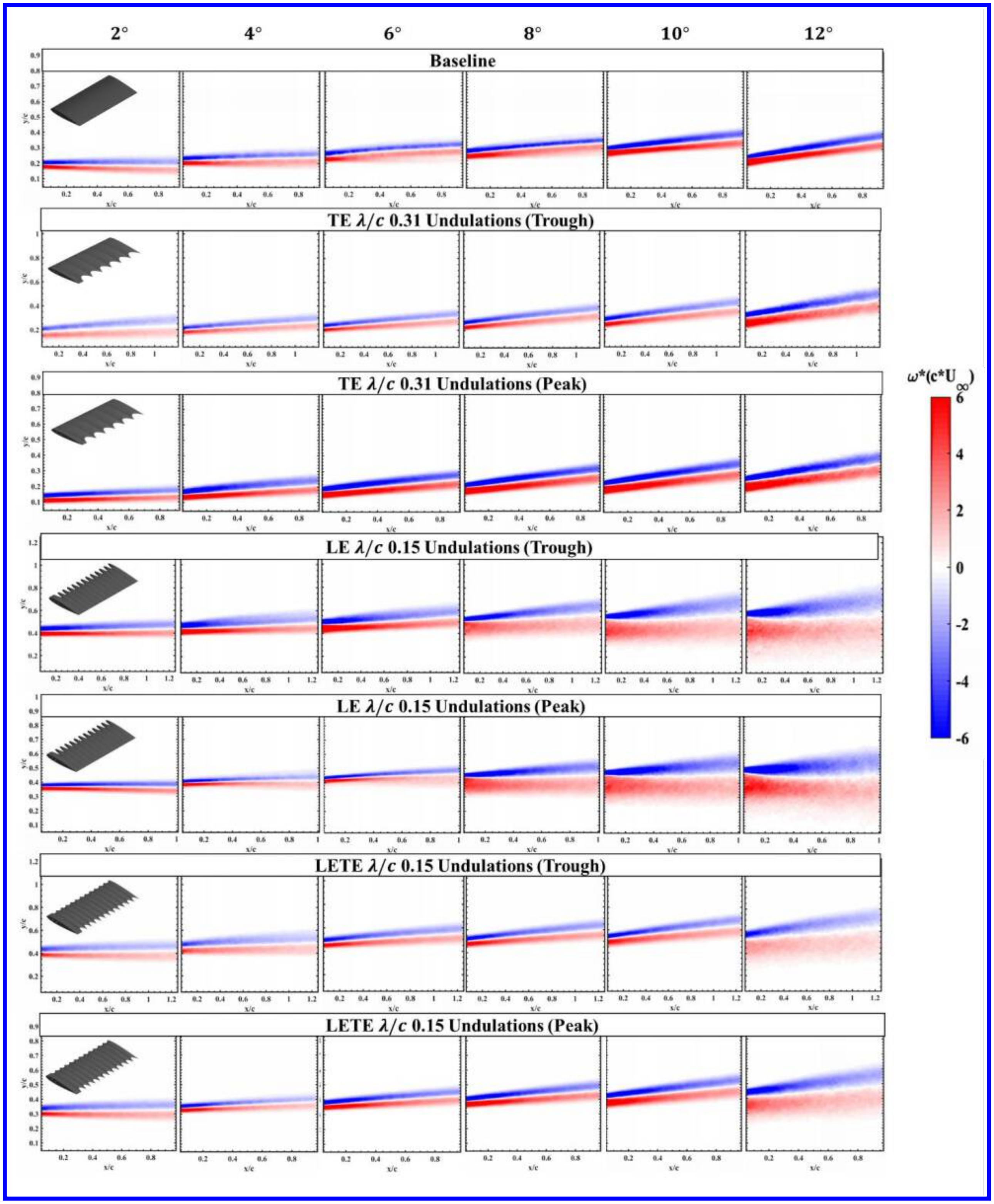

Fig. 11 Vorticity contours with increasing angle of attack for baseline, TE undulated wing, LE undulated wing, and LETE undulated wing

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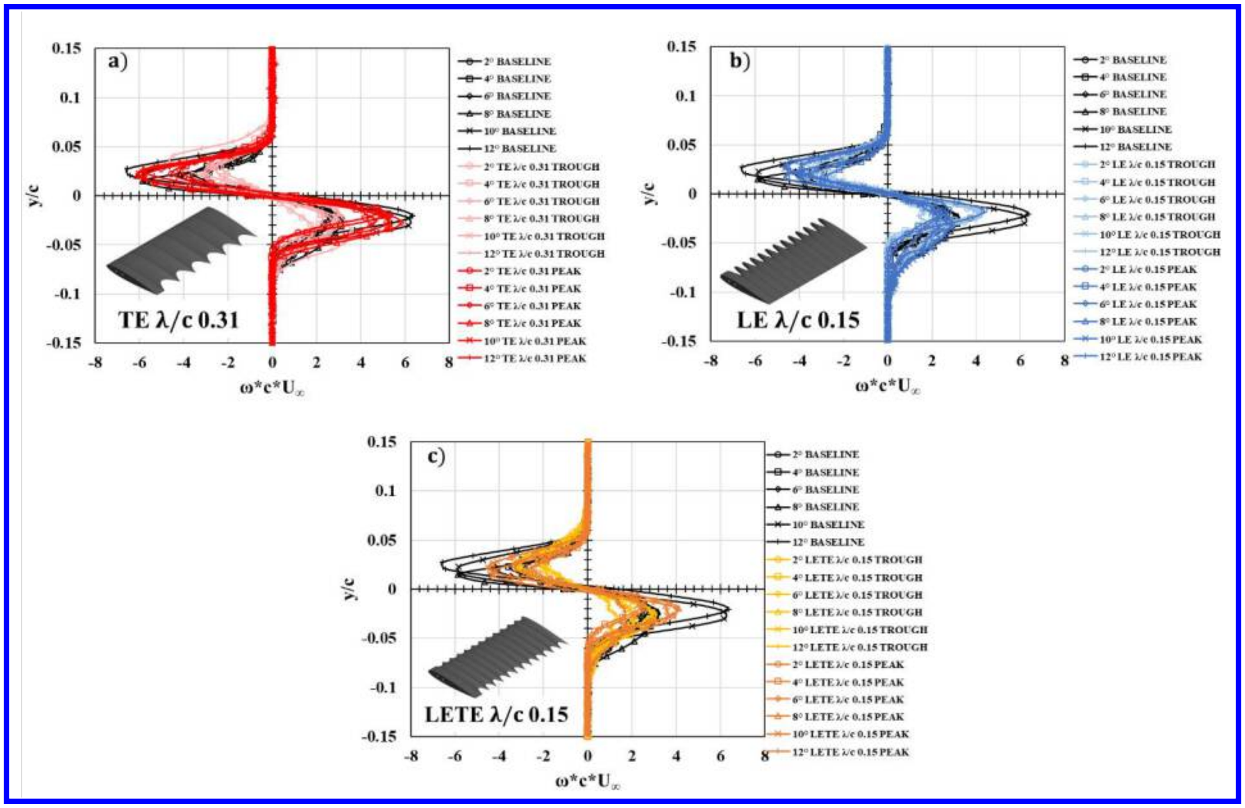

Fig. 12 Vorticity profiles with increasing angle of attack for a) TE λ/c 0.31 case compared to baseline, b) LE λ/c 0.15 case compared to baseline and c) LETE λ/c 0.15 case compared to baseline

1. Wake Turbulence Intensity

The wake turbulence intensity (TI) is determined by

$$k = \sqrt{\frac{1}{2}(u'^2 + v'^2)} \tag{2}$$

where u' and v' are the fluctuating velocity components obtained through Reynolds decomposition. The freestream normalized turbulence intensity contours and profiles are shown in Fig. 13 and Fig. 14 respectively. The turbulence intensity is an indicator of the level of turbulence in the wake. By comparing the TI for the different cases, the effect of the undulations on the wake turbulence can be extracted which also has implications on the noise reduction. Except for the LE undulated wing case, it can be seen that the difference between the TI in the peak and trough plane is minimal. Even though the mean velocity and vorticity showed changes in the peak and trough plane, the TI shows very minimal differences. This indicates that unlike the mean velocity components, the fluctuating velocity components might be uniformly distributed in the near wake of TE and LETE undulated wing cases. The LE undulated wing, on the other hand, shows larger TI in the trough plane when compared to the peak plane. Once again, this is hypothesized because of the formation of counter-rotating streamwise vortices causing a secondary flow in the trough plane. The flow in the trough plane in the LE undulated wing case could be highly three-dimensional when compared to the other cases, causing an increased TI magnitude. The effect of TE undulations on the wake TI can be seen clearly where the magnitude of the TI is significantly reduced in the near wake when compared to the baseline across all angles of attack. The magnitude of TI however is greater than the baseline at 12° angle of attack in the TE undulated wing case indicating higher levels of turbulence. The LE undulated wing produces higher TI magnitude, especially beyond 8° angle of attack in both the peak and trough planes. This coincides with the aerodynamic efficiency trends seen in Fig. 1 where efficiency of the LE undulated wing abruptly decreases after 6° angle of attack. The LETE undulated wing, on the other hand, delays the separation until 12° angle of attack where higher TI magnitude is seen.

This provides additional evidence that the LETE wing exhibits characteristics of both the TE and LE undulated wings.

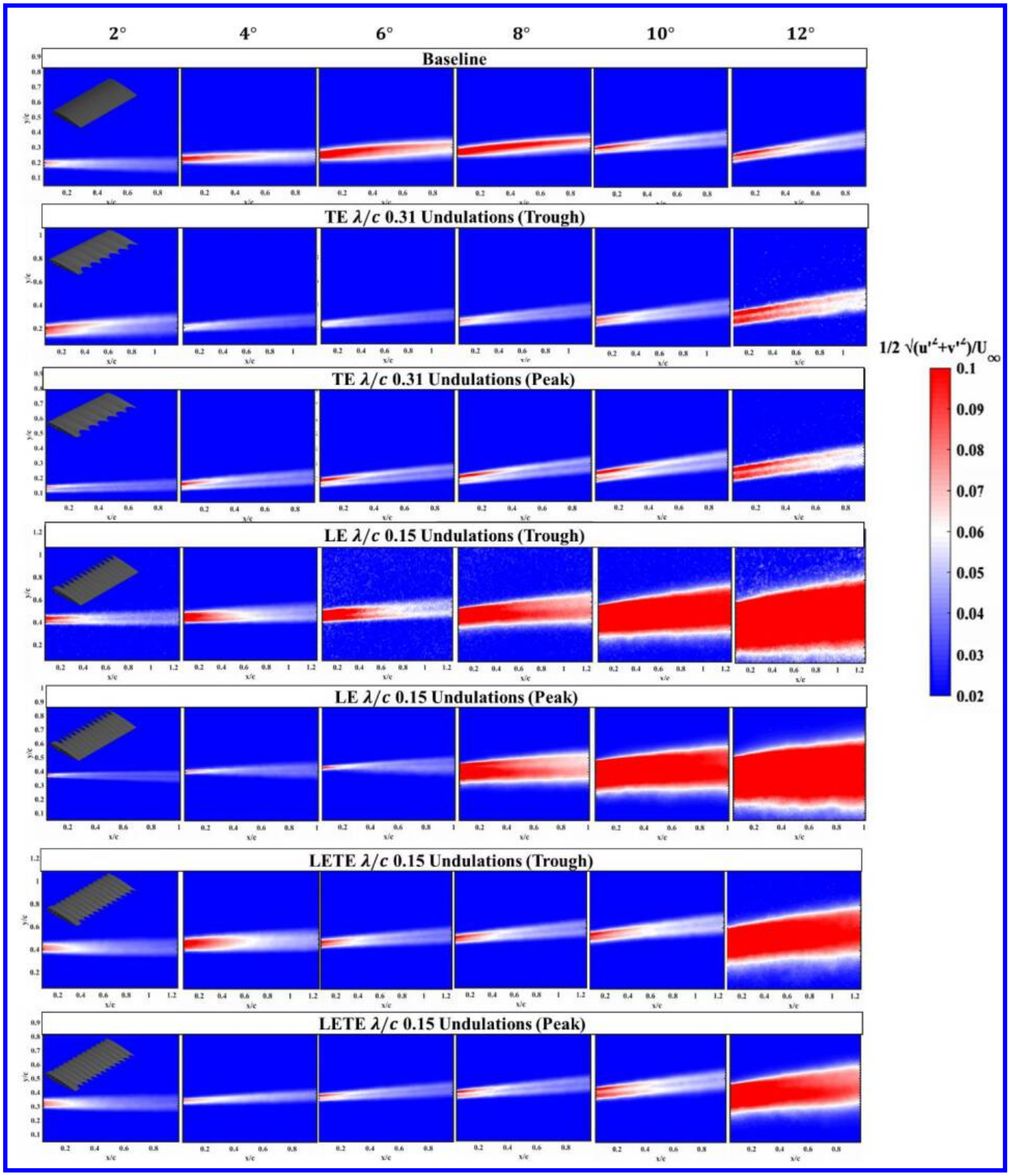

Fig. 13 Near wake normalized turbulence intensity contours with increasing angle of attack for baseline, TE undulated wing, LE undulated wing, and LETE undulated wing

The TI profiles shown in Fig. 14 highlight the TI trends and magnitudes seen in the contours. It should be noted that the TI profile switches from a bimodal distribution to a skewed distribution as a function of angle of attack in the baseline case. The undulations, however, changes this behavior and a bimodal distribution is seen across all cases. The TE undulated wing shows a reduced TI magnitude when compared to the baseline, but it also shows that

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the bimodal distribution of TI is maintained even at higher angles of attack when compared to the baseline. The LE undulated wing shows a significant increment in TI magnitude across all angles of attack and as mentioned earlier, maintains the bimodal distribution of the TI profile. Except for at 12° angle of attack, the LETE case shows similar behavior to the TE undulated wing case where the TI magnitude is lower than that of the baseline across all angles of attack.

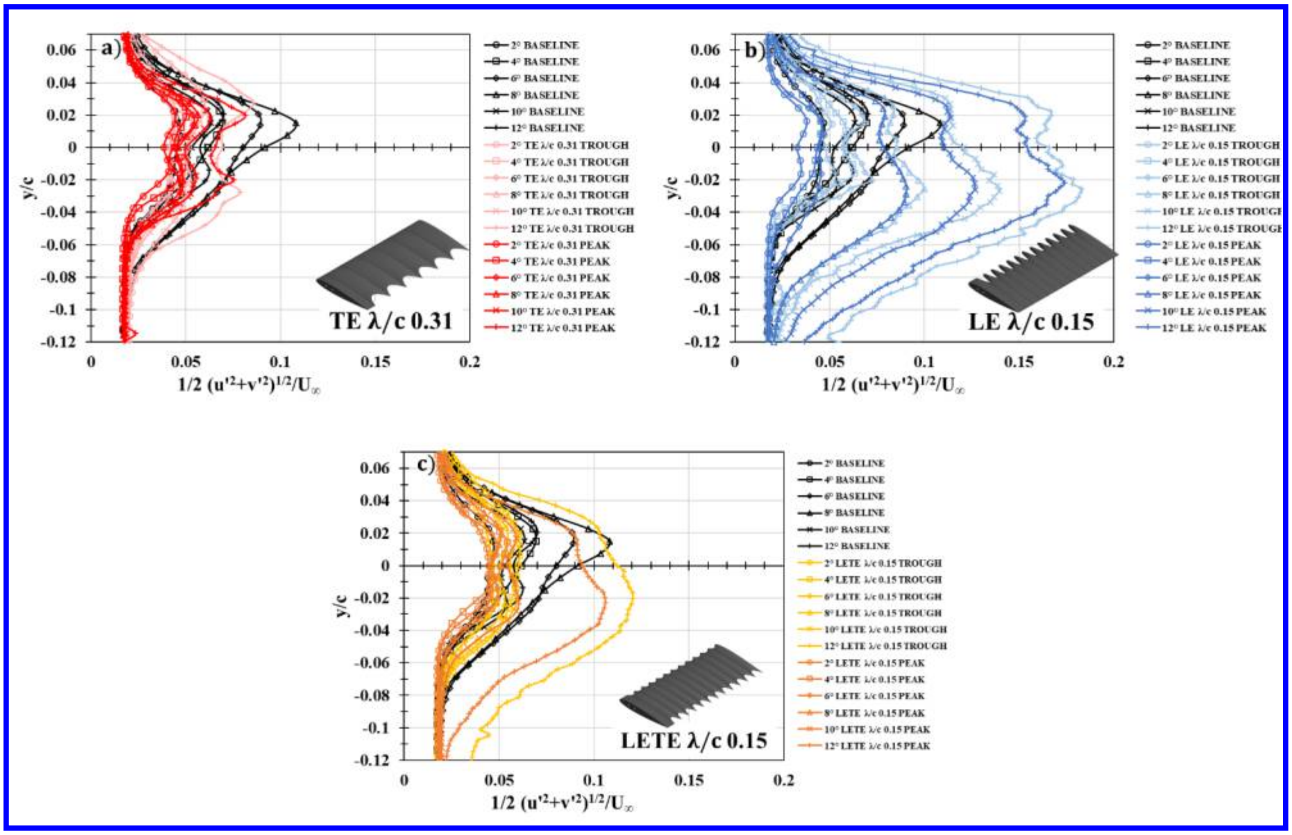

Fig. 14 TI profiles with increasing angle of attack for a) TE λ/c 0.31 case compared to baseline, b) LE λ/c 0.15 case compared to baseline and c) LETE λ/c 0.15 case compared to baseline

2. Reynolds Stress

The Reynolds shear stress quantifies the overall unsteadiness in the flow while also offering insight into surface pressure fluctuations. Results for Reynolds stress are shown using contours in Fig. 15 and profiles in Fig. 16. The trends observed in the TI contours are repeated once again in the normalized Reynolds stress contours. While the peak plane of the TE undulated wing show similar Reynolds stress to the baseline, the magnitude of the Reynolds stress is significantly reduced in the trough plane. Remembering the trends seen in other parameters, the trough plane of the LE wing shows much higher Reynolds stress than the baseline, especially past 6°. The LETE wing case shows the characteristics of both the TE and LE undulated wings.

Normalized Reynolds stress profiles can be seen in Fig. 16. The TE undulated case shows similar behavior to the baseline, and large differences are only seen at higher angles of attack. The case that deviates most significantly from the baseline is clearly the LE case as shown in Fig. 14b, where, a significant increase in Reynolds stress magnitude occurs at 8° for both the trough plane and peak plane of the LE wing, which is likely due to the onset of flow separation. Figures 14b-c also help to illustrate the asymmetry that is present between the upper and lower surface. The asymmetry continues to become more exaggerated as angle of attack increases. A more symmetric distribution exists at lower angles of attack.

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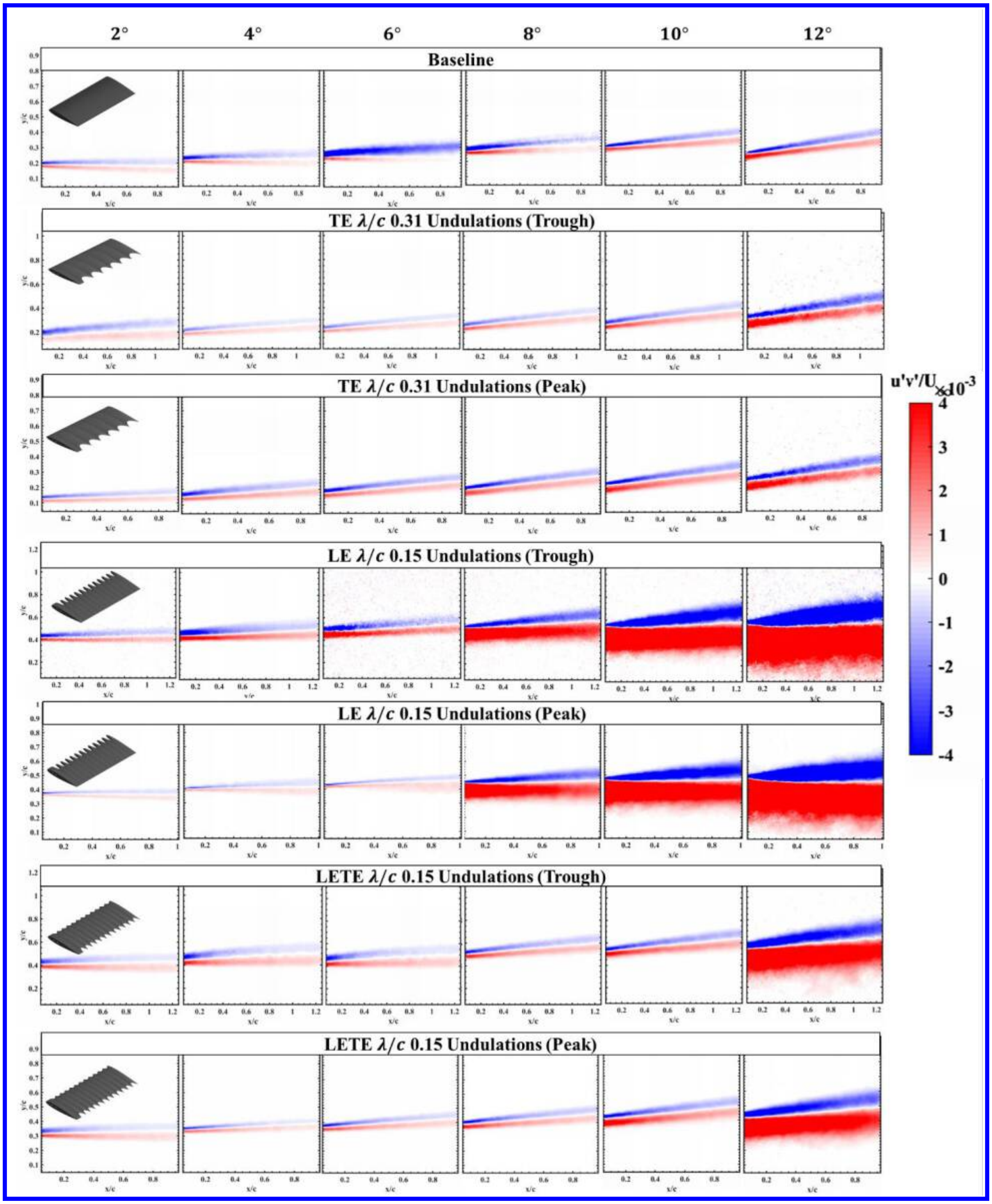

Fig. 15 Reynolds stress contours with increasing angle of attack for baseline, TE undulated wing, LE undulated wing, and LETE undulated wing

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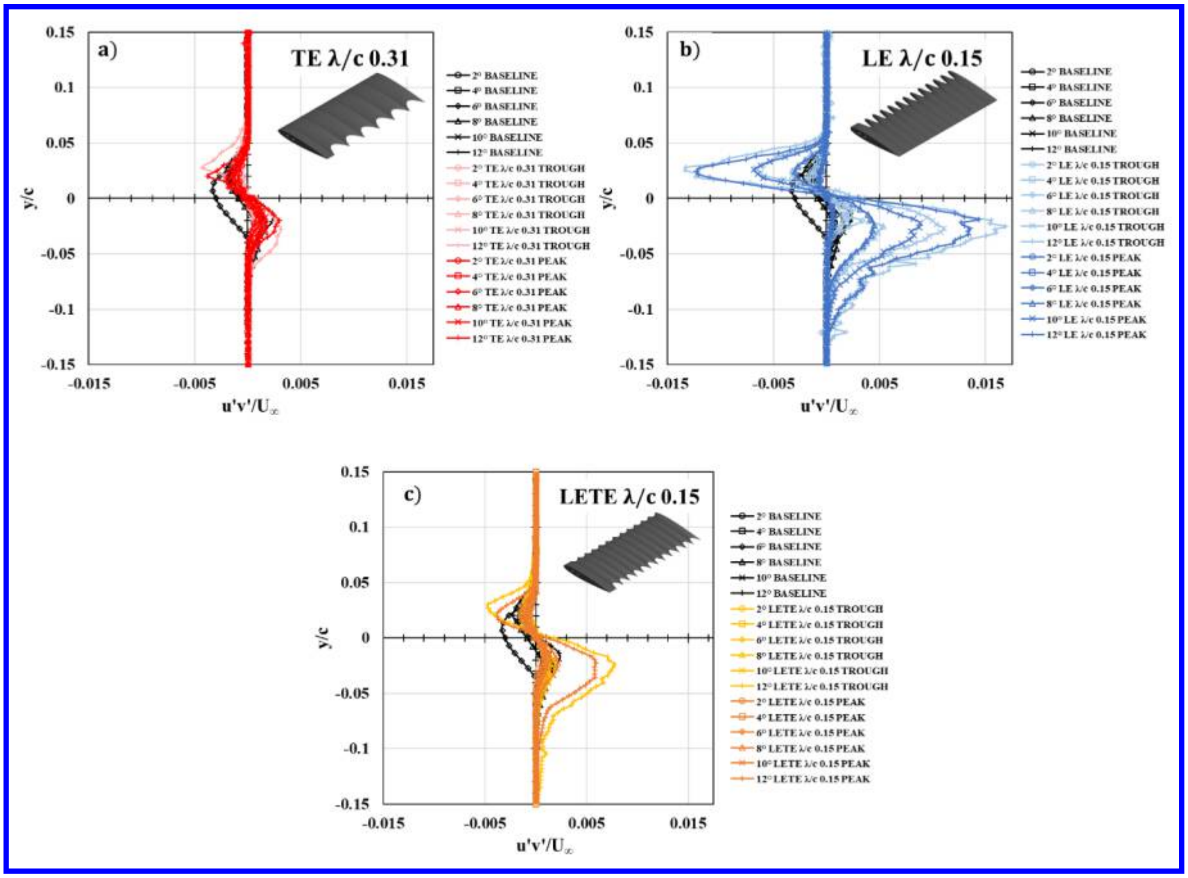

Fig. 16 Reynolds stress profiles with increasing angle of attack for a) TE λ/c 0.31 case compared to baseline, b) LE λ/c 0.15 case compared to baseline and c) LETE λ/c 0.15 case compared to baseline

C. Two-Point Correlation

Since the TE undulated wing shows drastic reductions in almost all flow properties for both mean and fluctuating velocity components, two-point correlations are performed in the near wake to provide greater insight into how the TE undulations affect the wake when compared to the baseline. It is well-known that vortex shedding frequency and turbulent length scales contribute to turbulence-induced pressure fluctuations, sound generation and structural vibrations. The two-point correlations can be used to gain insight into the shedding frequency and turbulent length scales by comparing changes in the coherent structures present in the wake between the baseline wing and wing with TE undulations. The two-point correlation also allows for the determination of length scales associated with the coherent turbulent motions. Bendat and Piersol [22] defined the two-point correlation as

$$\rho_{u_i u_j}(X_1, X_2, \tau) = \frac{\left[u_i'(X_1, t) * u_j'(X_2, t+\tau)\right]}{\sqrt{\overline{u_i'(X_1)^2}}\sqrt{\overline{u_j'(X_2)^2}}} \tag{3}$$

where X_1 and X_2 are two spatial locations in the PIV field of view, τ is the time delay (which is chosen to be zero for the results shown below), and u' represents the fluctuating velocities in i and j-directions. Fig. 17 and Fig. 18 show the contour levels of the normalized two-point correlation functions with zero time delay of the streamwise (u') and transverse (v') fluctuating velocities respectively for different angles of attack in the near wake of the baseline and TE undulated wing cases. In each case, the reference point (X_1) is chosen to be at the center of the upper shear layer as indicated in Fig. 17. The intent is to highlight the correlation in velocity fluctuations between the boundary layer flow from the upper surface and the near wake.

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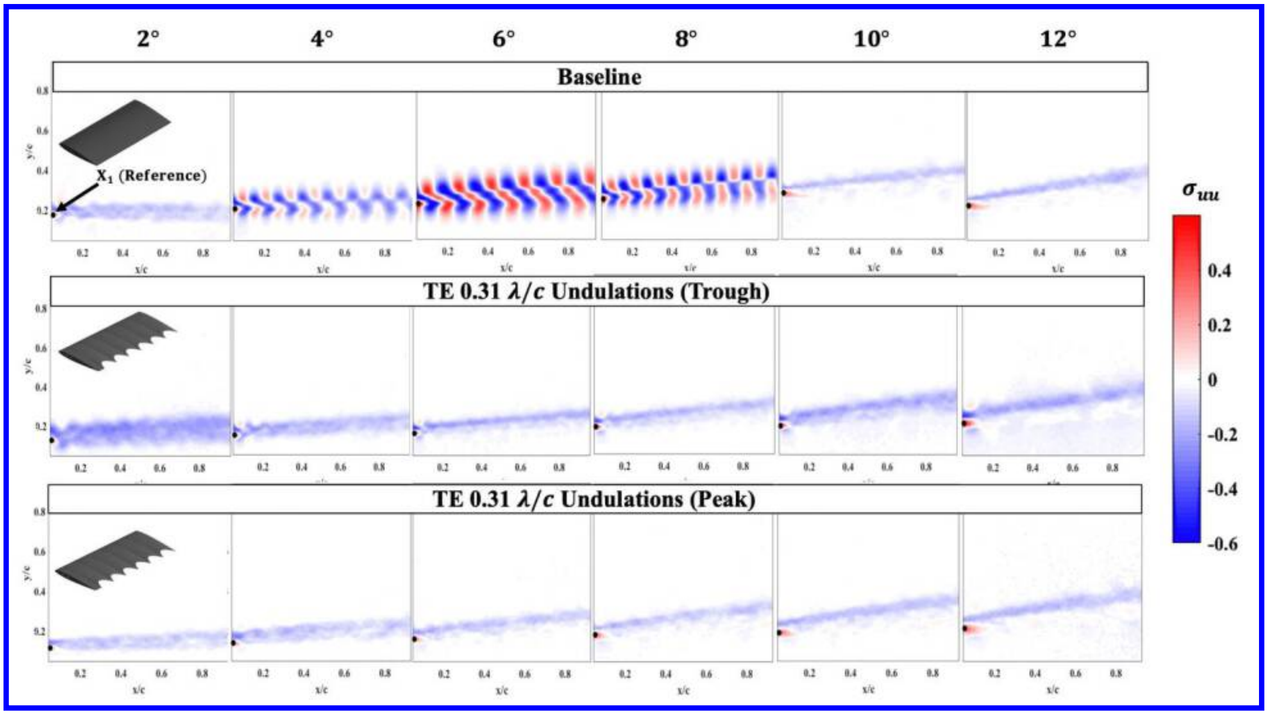

Fig. 17 ρ_{uu} correlation contours in the near wake of baseline and TE undulated wings at different angles of attack

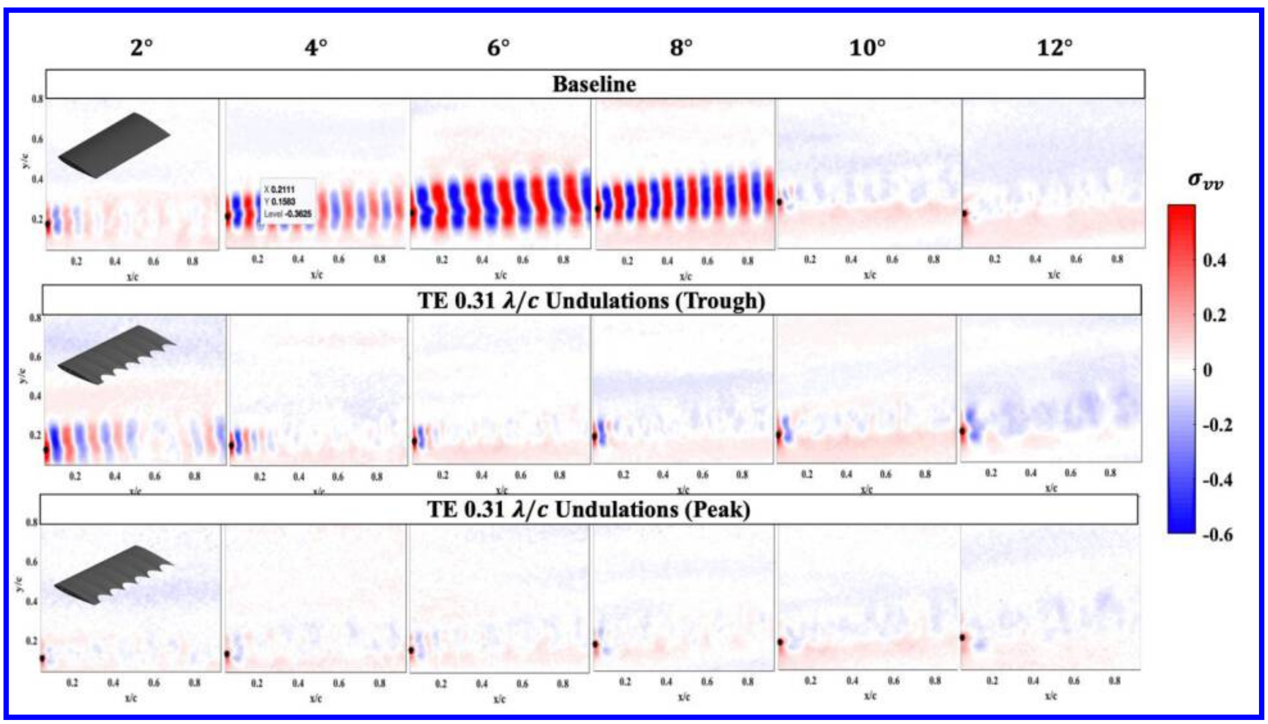

Fig. 18 ρ_{vv} correlation contours in the near wake of baseline and TE undulated wings at different angles of attack

Both the ρ_{uu} and ρ_{vv} contour images for the baseline case shows extensive coherent structures of alternating positive and negative correlation values. Specifically, spatially alternating regions of positive and negative correlation are indicative of the spatially and temporally periodic motions of the fluid. These motions can be related to the tonal character of fluctuations in the flow field at the frequency of vortex shedding. In the baseline case, the coherent structures are well formed in the shear layer emanating from the upper and lower surface of the wing, especially at 4°, 6° and at 8° angles of attack. As the angle of attack increases, the length scales (represented by the horizontal distance of each coherent structure) decrease due to increased vortex shedding frequency. This can be observed by quantifying the number of coherent structures in the wake. However, at higher angles of attack, at 10° the period shedding of the coherent structures disappears due to onset of flow separation. It is interesting to note that the TE undulations disrupt the periodic shedding of the coherent structures as no coherent structures are present in the near wake of the TE undulated wing in both the peak and the trough plane. The TE undulated wing creates almost uniform velocity correlations across the range of angles of attack. This is one of the major reasons why the TE undulated wings show a remarkable reduction in mean velocity, vorticity, TI and the Reynolds stress when compared to the baseline. The ρ_{vv} contours shows very subtle presence of the coherent structures, but they are not as clearly formed as seen in the baseline case.

IV. Conclusions

Investigations on wings with undulations along the leading edge, trailing edge, and both leading and trailing edge were conducted to validate the variation of aerodynamic coefficients seen in the previous study. Some major conclusions are as follows:

- In all mean flow properties such as the U-velocity and vorticity, the TE undulated wing case showed a remarkable reduction in wake thickness, momentum deficit, and vorticity magnitude in the trough plane when compared to the baseline across all angles of attack below 12°. The LE undulated wing, on the other hand, seems to have the opposite effect where mean flow properties were increased when compared to the baseline.
- The addition of TE undulations to the LE undulated wing (i.e. the LETE wing case) showed appreciable changes in wake characteristics where the abrupt flow separation seen in LE undulated wings was delayed until higher angles of attack. The LETE undulated wing showed characteristics of both the LE and TE undulated wings.
- The TE undulated wing showed a greater reduction in both TI and Reynolds stress when compared to the baseline and the other undulated wing cases.
- The two-point correlations revealed that reduction in the TI and Reynolds stress by the TE undulations is due to the successful disruption of the shedding of the coherent structures from the trailing edge of the wing.

All the trends seen in the free shear layer mimic the trends seen in the force-based experimental results seen in the previous study. While the TE undulated wing shows better performance at lower angles of attack, little to no differences were observed at higher angles of attack. The LE undulated wing shows a significant drop in efficiency as informed by the force-based testing results and the mean and fluctuating velocity components. However, they offer better post-stall characteristics. The wings with both TE and LE undulations exhibited the advantages of both LE-only and TE-only undulations where better performance is seen at lower angles of attack, mimicking the TE undulated wing performance as well as offer similar post-stall characteristics as the LE undulated wings.

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